

HEALTHCONNECT EXPERIENCE LAB

HealthConnect Analytics Testing & Refinement Report/Notebook

Week 7 — Analytical Testing, KPI Validation & Dashboard Refinement

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Track	Data Analytics
Programme	AnalystLab Africa — Premium Data Analytics Internship
Project	HealthConnect Experience Lab
Validation status	WEEK 7 PASS READY FOR WEEK 8 INTEGRATION

Testing → Finding → Action → Retest → Validated Improvement

1. Executive Summary

Week 7 focused on testing, refinement and end-to-end validation of the HealthConnect Data Analytics outputs developed through Week 6. Rather than introducing a new analysis, the work independently revalidated core KPIs, challenged the analytical evidence hierarchy, tested dashboard calculations and filter behaviour, reassessed business recommendations, and completed meaningful cross-track validation with Data Science.

Testing confirmed an Overall No-Show Rate of 51.15% within the attendance-observable population and Reminder Coverage of 72.68% across all appointments. Booking Lead Time remained the strongest standalone analytical differentiation of No-Show behaviour, with a 24.82 percentage-point gap between appointments booked 0–30 days and 31–60 days in advance. Recorded Previous No-Shows remained a complementary prioritisation signal, while Reminder Status and Distance to Clinic remained secondary or contextual factors.

Week 7 also identified and corrected a dashboard interpretation risk: interactive subgroup filtering could coexist with static reference-population conclusions. The final architecture therefore distinguishes MONITOR as an interactive operational view from DIAGNOSE and PRIORITIZE as validated analytical reference views. Recommendations were further refined into an explicit Decision → Action → Monitoring KPI → Evaluation → Extend / Adjust / Stop cycle.

Cross-track testing with Data Science produced two concrete modelling decisions. The engineered long_lead_followup feature was removed after failing to demonstrate meaningful incremental out-of-sample predictive value, while distance_to_clinic_km was retained after showing a small but consistent contribution within the multivariable model. These results reinforced the distinction between standalone analytical differentiation and multivariable predictive contribution.

Following end-to-end validation, no critical unresolved Analytics inconsistency or open Week 7 cross-track testing dependency remained. The HealthConnect Analytics workstream is therefore ready for Week 8 integration, subject to the documented data, observational-design and implementation limitations.

FINAL WEEK 7 ANALYTICS STATUS: PASS | READY FOR WEEK 8 INTEGRATION

2. Week 6 → Week 7 Transition

2.1 Week 6 baseline

- Validated analytical population and core KPI framework.
- Booking Lead Time identified as the primary analytical signal.
- Recorded Previous No-Shows identified as complementary evidence.
- Reminder Status and Distance retained as secondary/contextual evidence.
- Three-page Power BI decision-support architecture: MONITOR → DIAGNOSE → PRIORITIZE.
- Decision-support matrix and business recommendations prepared for further validation.
- Analytics findings handed off to Data Science where they could influence modelling decisions.

2.2 Week 7 validation objective

REVIEW → TEST → COMPARE → VALIDATE → REFINE IF NEEDED → RE-TEST → DOCUMENT

3. Testing Approach

Week 7 testing was designed to challenge Week 6 outputs rather than assume they were correct. The validation framework combined independent recalculation, reconciliation, robustness testing, dashboard scenario testing, recommendation reassessment, cross-track testing and retesting after refinement.

Independent recalculation: Recompute core metrics from source data using validated populations and denominators.

Reconciliation: Compare analytical calculations with Power BI measures and displayed values.

Robustness testing: Challenge important findings across segments and alternative specifications.

Control testing: Assess contextual variables that could potentially change the evidence hierarchy.

Dashboard scenario testing: Apply operational filters and test whether displayed messages remain contextually valid.

Recommendation testing: Check whether proposed actions remain proportional to evidence strength.

Cross-track testing: Use Analytics findings to challenge Data Science feature decisions through controlled testing.

Retesting: Re-test every genuine issue after refinement before assigning a final PASS.

VALIDATION CHAIN: TEST → FINDING → ACTION → RETEST → VALIDATED IMPROVEMENT

4. Testing Scope & Validation Record

Test	Component	Outcome
T01	Overall No-Show Rate	PASS
T02	Reminder Coverage	PASS
T03	Booking Lead Time	PASS
T04	Recorded Previous No-Shows	PASS
T05	Combined prioritisation matrix	PASS
T06	Dashboard values and measures	PASS
T07	Filter behaviour	REFINED + PASS
T08	Dashboard interpretation	REFINED + PASS
T09	Reminder Status	PASS
T10	Distance to Clinic	PASS
T11	Appointment Type × Lead Time	PASS
T12	Recommendations	REFINED + PASS
T13	Control variables	PASS
T14	Reminder Channel	PASS

Week 7 therefore produced 14 documented Analytics tests, including three areas requiring meaningful refinement followed by successful retesting.

5. KPI Validation

KPI	Validated definition / population	Result
Total Appointments	Full dataset	5,000
Attendance-Observable Appointments	Attended + No-Show	4,737
Overall No-Show Rate	2,423 / 4,737	51.15%
Reminder Coverage	3,634 / 5,000	72.68%

Core KPI definitions, denominators and dashboard values were independently revalidated and reconciled. No KPI inconsistency remained after Week 7 testing.

6. Analytical Validation Results

6.1 Primary evidence — Booking Lead Time

Booking Lead Time	No-Show Rate
0–7 days	29.47%
8–14 days	35.19%
15–30 days	45.53%
31–45 days	57.03%
46–60 days	71.36%

Consolidated comparison: 0–30 days = 39.13%; 31–60 days = 63.95%; gap = +24.82 percentage points. Additional validation produced Spearman $\rho \approx 0.287$ and an odds ratio of approximately 1.42 per additional 10 booking-lead days.

FINAL CLASSIFICATION: PRIMARY

6.2 Supporting evidence — Recorded Previous No-Shows

Appointments with 0 recorded previous no-shows had a 46.30% No-Show Rate, compared with 57.83% for appointments with at least one recorded previous no-show, a +11.53 percentage-point difference.

FINAL CLASSIFICATION: SUPPORTING / COMPLEMENTARY

6.3 Secondary / contextual evidence

Reminder Status showed a limited observational difference: 54.63% No-Show Rate without a reminder versus 49.86% with a reminder (+4.78pp for no reminder), with a very small effect size ($\Phi \approx 0.042$). Distance to Clinic also showed a very weak standalone association with No-Show behaviour (Spearman $\rho \approx 0.055$).

FINAL CLASSIFICATION: SECONDARY / CONTEXTUAL

6.4 Control variables

Appointment Day, Appointment Time, Waiting Time, Age Group, Gender, Previous Appointments and Reminder Channel did not provide sufficient standalone decision value to change the final evidence hierarchy.

7. Robustness & Segment Testing

Appointment Type	Lead-Time NSR Gap (0–30 vs 31–60 days)
Diagnostic	+22.24pp
Follow-up	+28.73pp
General	+23.51pp
Specialist	+23.16pp

Although Follow-up appointments showed the largest descriptive gap, the Lead Time × Appointment Type interaction was not statistically supported ($p \approx 0.326$). The Booking Lead Time pattern therefore remained robust across appointment types without supporting a separate Follow-up-specific analytical rule.

8. Dashboard Testing Evidence

8.1 Test scenario

Operational filters were applied across MONITOR, DIAGNOSE and PRIORITIZE to test whether visual values and explanatory messages remained aligned with the population being displayed.

8.2 Issue identified

Interactive filters changed subgroup values while some static analytical statements continued to represent the global validated population. This created a genuine interpretation risk: a user could read a global conclusion as if it described the currently filtered subgroup.

8.3 Refinement

Page	Final role	Interaction rule
01 MONITOR	Operational performance exploration	Interactive
02 DIAGNOSE	Validated analytical evidence	Reference analytical population
03 PRIORITIZE	Validated decision support	Reference analytical population

Explicit page guidance was added, misleading static elements were removed from the interactive page, and the final architecture was retested.

DASHBOARD RETEST: PASS

9. Before / After Improvements

Area	Before Week 7 refinement	Week 7 action	Validated after
Dashboard filters	Filtering could coexist with global conclusions	Tested context behaviour and separated page roles	Interactive vs validated views explicit
Decision signal	Static global signal on interactive page	Removed	No misleading static signal
Reminder gap	Static comparison vulnerable to filter context	Removed from interactive context	Context-consistent MONITOR
Recommendations	Action-oriented recommendations	Added explicit evaluation loop	Decision → Action → KPI → Evaluation → Extend/Adjust/Stop
Lead × Appointment Type	Descriptive Follow-up gap could attract overinterpretation	Interaction formally tested	No separate Follow-up rule
long_lead_followup	Candidate Data Science feature	Cross-track ablation test	Removed
Distance	Weak standalone Analytics signal	Cross-track multivariable test	Retained only as model feature; Analytics classification unchanged

10. Refined Dashboard / Visualisations

The final Power BI dashboard contains three validated pages. The final screenshots are retained as supporting evidence in the Week 7 submission package; the functional roles are summarised below.

01 — MONITOR | Performance Overview

Monitor overall appointment outcomes and explore operational contexts.

02 — DIAGNOSE | No-Show Differentiation

Communicate the validated analytical evidence hierarchy.

03 — PRIORITIZE | Attention Priorities

Translate validated evidence into group-level decision support.

The dashboard was frozen after visual review, calculation reconciliation, filter testing and retesting.

11. Validated Key Findings

Finding 1: Booking Lead Time remains the strongest standalone differentiation of No-Show behaviour.

Finding 2: Recorded Previous No-Shows provides complementary information and strengthens group-level prioritisation when combined with Booking Lead Time.

Finding 3: Reminder Status shows only weak observational differentiation and does not establish reminder effectiveness.

Finding 4: Distance has weak standalone Analytics differentiation and should not drive standalone operational targeting.

Finding 5: The Lead-Time pattern persists across Appointment Types, but there is insufficient evidence for a distinct Appointment-Type interaction or Follow-up-specific rule.

12. Updated Business Insights

Week 7 validation confirmed that Booking Lead Time remains HealthConnect's strongest standalone analytical differentiation of No-Show behaviour, while Recorded Previous No-Shows provides complementary prioritisation information. The combined evidence supports group-level prioritisation for further attendance-support evaluation rather than deterministic individual prediction. Reminder Status and Distance remain secondary or contextual factors and do not independently justify operational targeting. Robustness testing further showed that the Lead-Time pattern persists across appointment types without establishing a distinct Appointment-Type interaction. These findings therefore strengthen the decision-support framework while preserving the observational, non-causal and data-quality guardrails established during the analysis.

13. Updated Recommendations

1 — PRIORITISE | Longer Booking Lead Times

Use longer-lead appointments as the primary group for further attendance-support evaluation.

Monitoring KPI: No-Show Rate within the evaluated longer-lead population.

2 — REFINE | Recorded Previous No-Shows

Use Recorded Previous No-Shows cautiously as complementary information when refining the population selected for evaluation.

Monitoring KPI: No-Show Rate by Recorded Previous No-Show status within the evaluated population.

3 — TEST & MEASURE | Attendance-Support Strategy

Test reminder, confirmation or other attendance-support interventions before wider deployment.

Evaluation principle: measurable improvement relative to a predefined reference.

VALIDATED EVIDENCE → DECISION → ACTION → MONITORING KPI → EVALUATION → EXTEND / ADJUST / STOP

14. Cross-Track Testing Contribution

Week 7 required Data Analytics findings that could influence model features or modelling decisions to be meaningfully tested with Data Science. The Analytics track provided validated evidence on Booking Lead Time, Recorded Previous No-Shows, Distance to Clinic and the Lead Time × Appointment Type interaction, together with two explicit model-testing requests.

14.1 CT01 — long_lead_followup

Data Science performed a with-vs-without ablation using the same Logistic Regression pipeline, patient-grouped split and preprocessing. The single-split AUC difference was +0.0017 with bootstrap 95% CI [-0.0008, 0.0041], while grouped 5-fold cross-validation showed a mean difference of +0.0009. No statistically meaningful incremental predictive value was demonstrated.

Action: Removed from the Week 8 candidate feature set.

14.2 CT02 — distance_to_clinic_km

The single-split bootstrap result was inconclusive, but grouped 5-fold cross-validation favoured the model with Distance in 5 out of 5 folds, with a mean AUC difference of +0.0043. Removing Distance also reduced the Top-20%-risk lift from 1.44× to 1.40×.

Action: Retained in the Data Science model as a modest multivariable contributor; its Analytics classification remains Secondary / Contextual.

CROSS-TRACK LEARNING: STANDALONE ANALYTICAL DIFFERENTIATION ≠ MULTIVARIABLE PREDICTIVE CONTRIBUTION ≠ CAUSALITY

15. Assumptions, Limitations, Risks & Dependencies

15.1 Key assumptions

- The appointment record remains the validated unit of analysis.
- Attendance analysis excludes Cancelled appointments; operational coverage metrics may use the full dataset.
- Recorded Previous No-Shows is interpreted as a recorded appointment-level attribute, not a reconstructed longitudinal patient history.

15.2 Accepted limitations

- Repeated patient identifiers do not support reliable reconstruction of complete longitudinal patient histories.
- Distance and Waiting Time contain a small proportion of missing observations.
- Some extreme segments have limited sample sizes and are interpreted cautiously.
- The dataset is observational; associations do not establish causal effects.
- The Power BI file remains editable, so validated page behaviour also depends on governance and documented usage.

15.3 Residual risk and dependencies

Intervention effectiveness remains unvalidated and must follow a Test & Measure approach. Both Week 7 Analytics → Data Science testing dependencies have been resolved; no open cross-track testing dependency remains for the Analytics workstream.

16. Remaining Analytical Issues

No critical unresolved Analytics inconsistency was identified after Week 7 testing and end-to-end validation.

- Longitudinal patient-history reliability remains limited.
- Evidence remains observational rather than causal.
- Sparse extreme segments require cautious interpretation.
- Attendance-support intervention effectiveness remains untested.
- Validated dashboard and interpretation governance must be preserved during final integration.

17. End-to-End Validation

End-to-End Layer	Final Result
Data → Population	PASS
Population → KPI	PASS
KPI → Findings	PASS
Findings → Evidence Hierarchy	PASS
Evidence → Business Insights	PASS
Insights → Decision Support	PASS
Decision Support → Dashboard	PASS
Dashboard → Recommendations	PASS
Recommendations → Evaluation Cycle	PASS
Analytics → Data Science	PASS
Cross-Track → Solution Refinement	PASS
Guardrails & Interpretation	PASS

The full chain Data → KPI → Findings → Insights → Decision Support → Dashboard → Recommendations → Cross-Track Integration was therefore validated without identifying a critical unresolved Analytics inconsistency.

18. Week 8 Readiness & Recommendations

18.1 What is now validated?

The core Analytics solution, KPI definitions, evidence hierarchy, dashboard architecture, decision-support framework, business recommendations and cross-track interpretation are validated for Week 8 integration.

18.2 What remains uncertain?

Intervention effectiveness remains the principal operational uncertainty. Structural uncertainty also remains around longitudinal patient-history reliability, sparse extreme segments and causal interpretation beyond the observed dataset.

18.3 What must be completed before final integration/presentation?

- Preserve the validated Analytics evidence hierarchy in final storytelling.
- Integrate Analytics and Data Science findings without conflating their evidence types.
- Maintain non-causal and non-deterministic interpretation guardrails.
- Present attendance-support interventions under a Test & Measure framework.
- Carry forward known limitations in executive communication.
- Ensure consistency across Notebook/Report, Dashboard, Project Summary, Cross-Track Evidence, GitHub/Drive and final presentation.

WEEK 8 READINESS: DATA ANALYTICS READY FOR INTEGRATION

19. Conclusion

Week 7 transformed the HealthConnect Analytics work from a set of plausible Week 6 outputs into a tested and traceable decision-support solution. Core KPIs and analytical findings were independently revalidated, genuine dashboard interpretation weaknesses were corrected and retested, recommendations were strengthened through an explicit evaluation cycle, and cross-track testing produced concrete modelling refinements. The final Analytics evidence hierarchy remains centred on Booking Lead Time, supported by Recorded Previous No-Shows, while weaker factors remain appropriately contextualised. With no critical unresolved Analytics inconsistency or open Week 7 cross-track dependency identified, the workstream is ready to proceed to Week 8 final integration while preserving the documented analytical and decision-making guardrails.

Appendix A — Evidence Reference

Detailed test-level evidence is maintained in the Week 7 Testing & Validation Record. Cross-track testing evidence is maintained in the HC-POD Cross-Track Testing & Validation Evidence document and the supporting Data Science notebook. The final Power BI file and dashboard screenshots form part of the supporting submission evidence.

Cross-track collaborator: AYDEN NGNINTEDEM DEMANOU — Data Science, HC-POD 01.